

CHIGNIK, AK, USA

WMO: 703333

Lat: 56.311N

Lon: 158.373W

Elev: 18

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 25518

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.6	12.3	-17.1	2.2	29.2	-11.4	3.0	36.2	34.9	17.0	32.6	18.3	18.9	270	0.748

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	63.9	55.5	61.6	54.2	59.4	53.4	57.4	62.2	55.7	60.4	54.3	58.3	9.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.7	63.8	59.3	53.8	61.6	58.1	52.2	58.0	56.8	24.6	62.2	23.6	60.1	22.8	58.6	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.9	25.9	23.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	32.0	34.9	29.8	36.0	43.3	49.0	53.3	54.8	50.5	43.0	36.8	31.9
	DBStd	10.42	8.37	6.46	9.02	5.54	4.86	3.48	2.96	3.19	3.48	5.35	6.31	8.05
	HDD50	3519	559	422	628	420	215	57	4	2	32	222	396	562
	HDD65	8654	1024	842	1091	870	674	481	364	318	434	683	846	1027
	CDD50	344	0	0	2	0	7	27	105	149	49	5	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	9.6	9.9	9.6	11.9	9.9	8.7	7.7	7.1	8.1	9.0	10.4	11.0	11.5
(15) Precipitation	PrecAvg	35.8	3.2	2.7	2.2	1.9	2.0	2.4	3.1	4.1	3.8	3.5	3.3	3.5
	PrecMax	47.0	10.9	6.3	4.6	3.6	3.9	3.8	5.3	6.1	5.9	6.3	5.6	7.0
	PrecMin	27.5	0.8	0.5	0.4	0.5	0.4	1.3	1.2	2.2	1.9	1.3	1.1	1.3
	PrecStd	5.1	1.7	1.2	1.1	0.7	0.9	0.7	0.9	0.9	0.9	1.2	1.2	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	45.5	57.1	49.9	63.9	68.4	66.3	67.7	62.6	55.4	52.0	46.5
		MCWB	39.4	40.7	45.9	42.1	50.7	56.9	56.4	60.3	54.8	50.3	47.3	41.9
	2%	DB	43.4	43.5	45.7	46.2	56.7	61.2	63.1	64.0	59.0	53.9	48.0	45.1
		MCWB	39.6	39.2	39.9	40.4	47.3	51.5	55.2	57.0	53.0	49.4	44.7	41.3
	5%	DB	42.7	42.8	42.9	44.9	52.5	56.9	60.8	62.5	56.9	52.2	46.1	43.2
		MCWB	39.2	38.8	38.9	39.6	45.5	49.5	54.1	55.9	51.5	48.2	42.7	39.4
	10%	DB	41.0	41.2	40.7	43.0	49.1	54.5	57.4	60.5	55.1	50.2	45.1	41.3
		MCWB	38.0	37.8	37.5	38.4	44.2	48.6	52.0	54.8	50.4	46.5	41.4	38.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	42.2	46.5	44.4	51.2	57.0	58.4	61.6	56.6	52.1	47.3	44.3
		MCDB	44.3	45.2	57.3	47.2	61.7	66.9	64.4	66.7	60.7	54.5	49.2	45.5
	2%	WB	40.8	40.9	42.0	42.0	48.4	52.7	56.2	58.3	54.1	50.3	45.7	41.8
		MCDB	43.1	43.5	45.5	45.1	55.9	59.2	61.5	62.9	57.5	52.7	47.1	44.6
	5%	WB	39.8	39.9	39.4	40.6	46.3	50.8	54.7	56.6	52.7	48.6	43.7	40.2
		MCDB	42.1	42.2	42.6	43.7	51.2	55.7	59.2	60.9	56.0	51.2	46.0	43.2
	10%	WB	38.6	38.8	37.6	39.3	44.9	49.4	53.4	55.2	51.2	46.5	41.4	38.8
		MCDB	40.8	41.1	40.6	42.3	48.4	53.9	56.9	59.3	54.8	49.1	44.7	41.4
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	6.9	8.0	7.9	8.5	8.4	7.9	7.7	7.6	6.8	6.8	6.9
		MCDBR	7.1	7.9	8.5	10.1	14.2	13.6	12.2	11.3	9.5	8.7	8.6	7.6
	5% WB	MCWBR	6.0	6.3	6.1	6.9	8.8	8.3	8.0	7.4	6.7	6.7	7.2	6.3
		MCWBR	6.3	7.0	7.9	8.5	11.8	11.9	10.9	10.0	9.1	8.0	8.5	7.0
(40) Clear-Sky Solar Irradiance	taub	taub	0.281	0.344	0.375	0.414	0.417	0.402	0.386	0.383	0.368	0.339	0.327	0.283
		taud	2.365	2.195	2.117	2.082	2.144	2.293	2.407	2.405	2.377	2.322	2.137	2.271
	Ebn at Noon	Ebn at Noon	203	220	243	252	260	267	269	262	248	222	165	164
		Edh at Noon	15	27	37	44	44	39	34	32	29	23	19	13
(44) All-Sky Solar Radiation	RadAvg	RadAvg	142	311	731	1092	1409	1508	1354	1137	798	436	189	101
	RadStd	RadStd	21	39	118	99	168	156	125	101	73	36	22	11

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page